Contents

1 Reviewer one **1**
1.1 Changes made 1

2 Reviewer two **2**
2.1 Changes made 2

3 reviewer three **3**
3.1 Changes made 5

4 reviewer four **5**
4.1 Changes made 6

1 Reviewer one

- The paper has no references, despite the fact that the author(s) are not the first or only researchers to use the MIDFIELD database, nor are they the first or only to consider the interdisciplinary nature of many engineering fields and the people who practice in those fields. A search on some relevant keywords in the ASEE PEER database brings up an enormous number of papers, and more are published in non-ASEE venues. The author(s) really need to set forth at least a few of the more recent works in this area, and then explain how their work adds to what is already there. I do believe it is of value, but it needs to be set into a larger context.
- In addition, the author(s) should check out the ASEE author guide, as their paper doesn't conform to standard ASEE format.

1.1 Changes made

- Added five references to `bibliography.bib` and cited them throughout the paper using org-cite with natbib/IEEEtranN export. Lord et al. (MIDFIELD as a resource) is cited in Application Spaces, Background, Methodology, and Data Source. Klein (interdisciplinarity theory) is cited in Study Motivation and Limitations. Marquardt et al. and Franco et al. (broadening participation, interdisciplinary initiatives) are cited in Study Motivation. Denning & Tedre (computational thinking) is cited in Background.

- The novelty of the multi-metric approach over prior single-metric (Jaccard) analyses is now stated explicitly in Data Analysis and the abstract.
- References section uses `#+print_bibliography:` to generate from bibtex.
- ASEE format compliance is noted and will be addressed in the final submission formatting pass.

2 Reviewer two

- Motivation/contribution needs to be stronger; back with evidence. Explicitly clarify novelty; define relationship labels (e.g., decision rules); is enrollment a proxy for relevance, and also requirements, transfer credit, etc.? In results, discuss the findings is a less speculative way.

2.1 Changes made

- Motivation now cites evidence from Franco et al. on broadening participation through interdisciplinary curricula and Marquardt et al. on interdisciplinary engagement.
- Novelty is now stated explicitly: prior analyses used only the Jaccard index on binary course data; our contribution is applying multiple complementary metrics to frequency-weighted vectors to distinguish relationship types (Data Analysis section).
- Relationship labels are now formally defined in the Application Spaces section with metric-based decision rules. Three categories are identified: "nested" (high Overlap Coefficient with low Jaccard, indicating subset containment), "parallel" (high Jaccard/Kulczynski with low TF-IDF, indicating a shared base with different specializations), and "convergent" (high scores across all metrics, indicating deep curricular alignment). These labels were chosen to describe the geometric relationship in the data without implying genealogical or disciplinary hierarchy.
- The Application Spaces section now acknowledges that enrollment patterns reflect degree requirements, prerequisite structures, and scheduling constraints in addition to student interest. Language throughout

now refers to "curricular overlap" rather than directly claiming interdisciplinary engagement.

- All numerical claims in the results have been verified against the computed JSON similarity matrices. Several values were corrected: Math-Physics Jaccard from ~36% to ~23%, Physics-MechEng Overlap from ~45% to ~65%, engineering TF-IDF median from ~7% to ~3%.
- Results sections reordered by priority of finding: Engineering (Aero-Industrial convergent pair), Sciences (Physiology-Environmental Design-Physics cluster), Computing (IT-CS nested pair), STEM Overall (Physics-MechEng nested pair).
- Sciences section rewritten: the strongest PMI cluster is Physiology-Environmental Design-Physics (not "environmental studies and physiology" as previously stated). Math-Physics PMI is low (~4%), contradicting the prior claim of a "strong connection." Math acts as a generalist connector; Physics as a specialist.

3 reviewer three

- This manuscript addresses a timely and relevant topic in engineering education, namely the identification of interdisciplinary relationships through analysis of course enrollment patterns. The motivation is well aligned with current discussions around convergence, cross-disciplinary collaboration, and curriculum design. The paper also demonstrates technical effort in applying multiple similarity metrics and visualizing relationships among programs. However, in its current form, the manuscript has several substantive issues that limit the strength of its research contribution. The comments below are intended to provide constructive guidance for strengthening the work.
- A primary concern is the clarity of the research contribution. While the manuscript presents an analytical approach for examining course overlap, it remains unclear what new educational understanding emerges from the analysis beyond identifying expected similarities among related programs. The paper effectively demonstrates that similarity metrics can be applied to curricular data, but it does not sufficiently establish how the results advance knowledge about interdisciplinarity, student behavior, or curriculum development. Strengthening the

connection between the analytical results and a clearly articulated educational insight would significantly improve the contribution of the work.

- Related to this issue is the reliance on course enrollment as a proxy for interdisciplinary engagement or student interest. The manuscript assumes that enrollment reflects intentional investment in multiple disciplinary areas; however, course selection is often influenced by degree requirements, scheduling constraints, prerequisite structures, or institutional policies. These factors introduce important confounds that are not currently addressed. As a result, interpretations that link enrollment overlap directly to interdisciplinarity or collaboration may be overstated. A more cautious framing, or additional justification for this assumption, would strengthen the validity of the conclusions.

- The methodological section provides detailed descriptions of similarity metrics, but the analytical rationale for selecting these methods is less clearly developed. The manuscript explains how metrics such as TF-IDF or PMI are computed, yet it is not always evident why these particular measures are best suited to answer the research questions posed. The work would benefit from a clearer explanation of how each metric contributes uniquely to understanding interdisciplinary relationships, and whether alternative or simpler approaches would produce comparable insights. Without this clarification, the methodology risks appearing as a technical exercise rather than a research-driven design.

- The results section is largely descriptive, and many of the reported relationships appear consistent with what would already be expected based on disciplinary proximity. While visualizations are informative, the interpretation remains relatively limited, and the manuscript does not clearly demonstrate that the proposed approach reveals relationships that were previously unknown or difficult to identify through conventional curricular analysis. Expanding the discussion to explain why the findings matter for decision-making or educational practice would help strengthen the impact of the results.

- Another important limitation is the absence of validation or evaluation of the analytical outcomes. The paper does not currently include expert verification, comparison with known interdisciplinary initiatives, or any form of external validation that would demonstrate that the detected relationships correspond to meaningful educational or institutional realities. Including such validation, even in a limited form, would substantially increase confidence in the approach.

- Conceptually, the paper also treats interdisciplinarity primarily as course overlap, which simplifies a complex educational construct. Interdisciplinary learning typically involves integration of knowledge, methods, or perspectives across domains, not only shared enrollment patterns. Clarifying this distinction and situating the analysis within a stronger conceptual definition of interdisciplinarity would improve theoretical grounding.

3.1 Changes made

- Several of these points overlap with concerns raised by Reviewers 1, 2, and 4 and are addressed through those revisions (citations, enrollment proxy framing, metric rationale, non-obvious findings).
- The Limitations subsection in the Discussion now explicitly distinguishes curricular overlap from interdisciplinary knowledge integration, citing Klein. Validation against known interdisciplinary programs is listed as planned future work in the Conclusion.
- Metric selection rationale has been strengthened: the methodology section is now titled "Metric Selection Rationale" and each metric includes an explicit analytical question it answers (e.g., Jaccard: "How much total curricular territory do two programs share?"). The introductory paragraph explains why a single metric is insufficient and how the suite as a whole separates curricular width from specialized identity.
- Results now foreground non-obvious findings: the Aerospace-Industrial convergent pair (highest similarity across every metric in engineering, ~5x the next TF-IDF pair) and the Physiology-Environmental Design PMI cluster (highest PMI in sciences at ~25%, connecting programs from different CIP families). Expected relationships (EE-CompE, Math as bridge) are presented as confirmatory context rather than primary findings.

4 reviewer four

- Why are there no citations or references to prior work?

- There are a few instances of sentences beginning with "This" that do not go on to describe what "This" is referring to. Example: "Students often have a level of agency in the courses they take on their paths to graduation, opting to take courses that may not be housed in their declared major's department. THIS [emphasis mine] indicates that the student has knowledge, skills, or attitudes (KSAs) from several disciplines and are interdisciplinary in their education." This what? Student agency, or this exposure to other subject areas?

- I like the WIP conclusions and plans for future work, though I would love to see more preliminary examples of novel relationships. Identifying that Software Engineering is effectively a subset of Computer Science, or that electrical engineering is very similar to Computer Engineering, are fairly intuitive relationships. However, an opportunity for an Aerospace/Industrial Systems engineering program could be deeply intriguing.

4.1 Changes made

- Citations added throughout; see Reviewer 1 response above.

- The ambiguous "This" in the Study Motivation has been rewritten. The sentence now reads: "Such cross-departmental enrollment indicates that students acquire knowledge, skills, or attitudes (KSAs) from several disciplines..." Other instances of dangling "This" throughout the paper have been revised to name their antecedent explicitly.

- The Aerospace-Industrial Engineering pairing is now the central finding of the Engineering results section. This pair is classified as "convergent" (a new third relationship category) because it dominates every metric: TF-IDF ~45%, Jaccard ~47%, Kulczynski ~66%, Overlap ~79%. The TF-IDF value is nearly five times higher than the next strongest engineering pair, making it a clear outlier. The Discussion uses this pairing as the primary example of an actionable insight (potential cross-disciplinary certificate or dual-major pathway).

- Results sections are now ordered by novelty: Engineering first (Aero-Industrial convergent finding), then Sciences (Physiology-Environmental Design cluster), then Computing (IT-CS nested relationship), then STEM Overall (confirmatory cross-domain patterns).